



สถาบันวิจัยเพื่อการพัฒนา (IRD)



จุฬาลงกรณ์มหาวิทยาลัย



มหาวิทยาลัยเทคโนโลยีราชมงคลล้านนา น่าน



วิทยาลัยชุมชนน่าน



มหาวิทยาลัยเกษตรศาสตร์

## High School Curriculum Development

### ADAPT Workshop Report

Satthasilaphet Rangarn School - February 12, 2026

This document reports about a curriculum planning workshop covering two integrated curricula — **Safe Vegetable Production** and **Indigenous Rice Farming**— developed collaboratively between secondary school teachers and university staff for implementation at Satthasilaphet Rangarn School, with practical links to the local community in Nan Province. First, we engaged students from Level 4 into a role-play game about chemical use in agriculture. They showed good understanding of the game and explored different scenarios with ease, which confirmed the interest of teachers in using simulation games as part of our curriculum. Based on the results of the previous workshop, held in December 2025, we further discussed key pedagogical orientations.

### Safe Vegetable Curriculum

**The learning units will be renamed to ensure consistency.**

**Units 4 and 5:** Focus on markets closer to school / markets near the school.

Visit Ban Na Kham Learning Center and invite resource persons: Ai Yai Ken Kakin Nan, Wat Pong Kham (in relation with Learning Unit 8) and Mobile Development Military Unit 31

### Unit 7 – Insects and Pests:

- Provide basic insect learning media for students, then have students go find insects in the greenhouse. The school's teachers lack expertise in insects, but university support would benefit students in future.
- Students' insect collecting will be linked with the biology teacher (Teacher Jongrak), learning basic safe management. Students should be able to recognize insects. For more academic content, students should learn about life cycles from the beginning.
- Aj. Buncha has expertise in plant diseases, fungi/spore analysis and diseases that arise (this is a request from Silaphet School — in game/media format). Suggest students collect fungi or soil samples for analysis and differentiate plant diseases (fungi, bacteria, viruses) specific to plants grown by students.
- Could place insect nets/jars at school for storing insects to ask teachers about, or photograph for analysis; create small insect collection booklets; the school should have a corner for

relevant teaching materials and specimen samples. (Aj. Buncha can teach how to make a demonstration kit.)

- Request a high-quality microscope, Aj. Buncha suggests using the microscope from Kasetsart University stored at the museum for the benefit of Silaphet students. Aj. Buncha will prepare a recorded presentation; in practical sessions, if budget allows, will bring students to stain and view fungi or bacteria.
- Teacher Jongruk proposes that when university lecturers come to teach, both M.4 and M.5 students will learn together on the relevant topics.

**Unit 8:** Aj. Priyanuth will support the fermented liquid fertilizer production topic.

**Unit 10:** Revise objectives to align better with marketing.

**In Term 2, Unit 8 – Food Processing:** Aj. Joy Suthida Panyain will support the training.

### **Curriculum Appropriateness Assessment**

When the curriculum will be completed, it will be compiled into a booklet. The school director, academic department head, and Saa School (Teacher Pop Pi) the curriculum head or school administrator — will evaluate the curriculum booklet no later than the end of April.

### **Indigenous Rice Farming**

**Unit 3 – Rice Variety Game:** focusing on rice characteristics or pure-bred rice seeds. Teaching media for students on GI rice, Nan rice varieties, and healthy rice as separate units. Planting by Panee method or throw planting (Teacher Jongruk already teaches this). For the **Rice Field Ecosystem**, Aj. Beer will consult the team and design a game consistent with the content.

When students go to the field, the main instructors will be local wisdom teachers, with schoolteachers assisting in video documentation. Hire a company to create storyboards divided into parts, assist teachers in oversight, and have the company handle editing.

## Safe Vegetable Curriculum - Term 1 (starts 11 May)

| #  | Topic                                                                                                             | Objectives                                                                                                                                                          | Methods                                                                                                                           | Assessment                                                                 | Equipment                                                   | Responsible                                                         |
|----|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------|
| 1  | <b>Agriculture Gen Z</b> - Food situation - Food security - Food safety - Agricultural ecology                    | Explain current food production situation, food security and food safety as related to livelihoods and occupation                                                   | Games, video clips, discussion of news/clips on food situation, discussion of community impacts, the meaning of "safe vegetables" | Discussion of problems, observation of behavior                            | Test worksheets, behavior observation forms                 | Aj. Phongchai, Teacher Sirilaksana                                  |
| 2  | <b>What are safe vegetables?</b> - Types of vegetables - Benefits - Local/indigenous vegetables - Growing methods | Explain principles and concepts of safe agriculture and agricultural ecology as appropriate career alternatives in the local community context                      | Group analysis of safe agriculture case studies, linking to local context, analyzing solutions                                    | Group presentations, assignments, teaching games                           | Tests/assignments, teaching games                           | Aj. Phongchai, Teacher Sairungg, Aj. Banchaa (Tropical Crop Center) |
| 3  | <b>How do vegetables grow?</b> - Physical-biological factors - Risk factors - Survey market demand                | Classify types of vegetables, their benefits, harvest age and period, including value of indigenous vegetables and local foods applicable to daily life and careers | Survey vegetables in community, classify types and benefits                                                                       | Practical/portfolio                                                        | Vegetable classification table, indigenous vegetable poster | Teacher Sairungg                                                    |
| 4  | - Market survey and data analysis                                                                                 | Explain and select appropriate vegetable growing methods suited to local resources and working costs                                                                | Survey form/report                                                                                                                |                                                                            |                                                             | Teacher Sairungg, Wat Pong Kham                                     |
| 5  | <b>Present survey results and plan production</b> - Select vegetable types - Production planning                  | Explain vegetable growth and analyze physical and biological factors affecting cultivation                                                                          | Presentation/Rubric                                                                                                               |                                                                            |                                                             | Teacher Sirilaksana                                                 |
| -  | <b>Mid-term Examination</b>                                                                                       |                                                                                                                                                                     |                                                                                                                                   |                                                                            |                                                             |                                                                     |
| 6  | <b>Prepare equipment, materials and seedlings</b> - Growing media/containers - Seedling nursery                   | Practice planting, caring for and managing safe vegetables including composting and biofertilizer correctly, economically and safely                                | Practical/observation                                                                                                             |                                                                            |                                                             | Teacher Phatnarin                                                   |
| 7  | <b>Pests and Insects</b> - Biological classification and management / AI                                          | Use biological methods to manage plant diseases, insects and pests, and apply digital/AI technology appropriately in agricultural work                              | Assignments/practical                                                                                                             | Insect identification game, specimen samples, reference books, video media | Teacher Sirilaksana, Aj. Banchaa, Dr. Kittiphum             |                                                                     |
| 8  | <b>Compost and Biological Fermentation</b> - Production and use of biofertilizer                                  | Plan safe vegetable production work from planting, care, harvest, storage, processing and sales as a livelihood                                                     | Actual work pieces                                                                                                                |                                                                            | Teacher Phatnarin, Aj. Piyanut                              |                                                                     |
| 9  | <b>Practice planting, caring and harvesting</b> - Actual planting - Care - Safe harvesting                        | Work collaboratively with responsibility, discipline, appreciation of honest work and environmental consciousness                                                   | Practical/learning journal                                                                                                        |                                                                            |                                                             | Teacher Phatnarin                                                   |
| 10 | <b>Marketing and Presenting Results</b> - Sales - Summary - Presentation                                          | Explain current food production situation, food security and food safety related to livelihoods and occupation                                                      | Presentation/self-peer assessment                                                                                                 |                                                                            |                                                             | Teacher Muktida                                                     |
| -  | <b>End-of-term Examination</b>                                                                                    |                                                                                                                                                                     |                                                                                                                                   |                                                                            |                                                             |                                                                     |

### Safe Vegetable Curriculum - Term 2 (starts 19 October)

| #  | Topic                                                                                                                                   | Objectives                                                                                                                                           | Methods                          | Assessment       | Tools                                       | Responsible |
|----|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|------------------|---------------------------------------------|-------------|
| 1  | <b>Importance of safe vegetables</b> - Food safety - Consumer health - Local resources                                                  | Explain the importance of growing pesticide-free vegetables and their impact on food safety, consumer health, and appropriate use of local resources | Test/discussion                  | Food safety game | Teacher Sairungg, Aj. Ornnadda              |             |
| 2  | <b>Survey community market demand</b> - Market survey - Data collection and analysis                                                    | Survey and analyze community market demand using systematic data collection                                                                          | Assignments/behavior observation |                  | Teacher Sairungg                            |             |
| 3  | <b>Present survey results and plan production</b> - Data presentation - Data-based production planning                                  | Present market survey results and use data to appropriately plan safe vegetable production                                                           | Assignments/tests                |                  | Teacher Sirilaksana                         |             |
| 4  | <b>Selecting short-cycle local crops</b> - Appropriate short-cycle crops such as sunflower sprouts, morning glory sprouts, bean sprouts | Plan and select appropriate short-cycle crops for local context such as sunflowers, morning glory sprouts and bean sprouts                           | Tests/analysis worksheets        |                  | Teacher Muktida, VCCD Nan, RMUT, PGS Nan    |             |
| 5  | <b>Preparing equipment and cultivation</b> - Equipment, materials, growing containers - Seedling nursery - Care                         | Practice preparing equipment, materials, growing containers, seedling nursery and correct and safe plant care                                        |                                  |                  | Teacher Muktida                             |             |
| -  | <b>Mid-term Examination</b>                                                                                                             |                                                                                                                                                      |                                  |                  |                                             |             |
| 6  | <b>Non-turning compost</b> - Production and use of environmentally friendly compost                                                     | Produce and use non-turning compost for environmentally friendly crop growing                                                                        |                                  |                  | Teacher Phatnarin                           |             |
| 7  | <b>Planning safe vegetable sales</b> - Sales planning linked to real situations                                                         | Plan sales and present a safe vegetable sales plan linked to real community situations                                                               |                                  |                  | Teacher Muktida                             |             |
| 8  | <b>Processing and local food from vegetables</b> - Food processing integrating local wisdom                                             | Process produce and prepare food from self-grown vegetables integrating local wisdom with community resource persons                                 |                                  |                  | Teacher Phatnarin, Ban Na Kham, Aj. Suthida |             |
| 9  | <b>Analyze production and sales results</b> - Analyze operating results, problems and improvement approaches                            | Analyze operating results and performance, reflect on problems and propose process improvements                                                      |                                  |                  | Teacher Sirilaksana                         |             |
| 10 | <b>Professional skills and teamwork</b> - Responsibility, communication, collaboration                                                  | Explain the importance of growing pesticide-free vegetables and their impact on food safety, consumer health and appropriate use of local resources  |                                  |                  | Teacher Sirilaksana, Aj. Ornnadda           |             |
| -  | <b>End-of-term Examination</b>                                                                                                          |                                                                                                                                                      |                                  |                  |                                             |             |

## Indigenous Rice Farming - Term 1

| # | Topic                                                                                                                                                                                                    | Outcomes                                                                                                                                            | Methods                                                                                                                                                                                                        | Assessment                                                                                                                                      | Tools                                                   | Responsible                                                       |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------------------------|
| 1 | <b>Sufficiency Economy Philosophy</b> - Meaning of Sufficiency Economy Philosophy - Components of the philosophy - Application approaches - Agricultural models under the Sufficiency Economy philosophy | Search for information, discuss and explain the Sufficiency Economy Philosophy principles and apply them to daily life                              | - Study video clips/songs/worksheets of H.M. King Rama IX's royal speeches - Group presentations of the Sufficiency Economy Philosophy applied to daily life at student and family level - Complete worksheets | 1. Check Worksheet 1 2. Observe activity process 3. Observe student behavior (desirable characteristics, competencies)                          | Worksheets 1–4 on the Sufficiency Economy Philosophy    | Teacher Jongruk, Iham Kham                                        |
| 2 | <b>Thai Proverbs, Sayings and Idioms about Rice and Rice Farming</b> - Thai proverbs, sayings and idioms about rice and rice farming in the local area and Thailand                                      | Search for information, discuss and explain proverbs, sayings and idioms about rice farming in local areas and Thailand                             | - PowerPoint presentations - Discussion of proverb examples - Group search and presentation - Design and present Thai proverbs about rice farming - Rice farming video clips                                   | 1. Check Worksheet 1 on proverbs/idioms with the word "rice" 2. Test on proverbs/idioms 3. Observe activity process 4. Observe student behavior | Assessment forms for worksheets and tests               | Teacher Jongruk, Iham Kham                                        |
| 3 | <b>Thai Rice Varieties – Cherish and Conserve</b> - Importance of Thai rice - Rice varieties - Health rice - GI rice - Glutinous rice - Non-glutinous rice                                               | Search for information, discuss and explain the importance of rice, types of rice varieties, health rice, GI rice, glutinous and non-glutinous rice | - Group search and presentation on importance of rice, rice varieties, health rice, GI rice - Study video clips/rice grain samples/photos - <b>[Rice variety card game (Thai rice knowledge)]</b>              | 1. Check Worksheet 1 on rice importance and types 2. Check Worksheet 2 on health and GI rice 3. Check test on diversity of Thai rice varieties  | Assessment forms for worksheets and tests; rice samples | Aj. Phongchai and team to design rice variety game                |
| 4 | <b>Rice Grains</b> - Grain structure - Seedling nursery - Grain germination - Seedling growth                                                                                                            | Survey, examine and explain rice grain structure, rice seed germination, and seedling growth                                                        | - Group presentations on rice grain structure - Research seedling/germination methods - Hands-on seedling practice                                                                                             | 1. Check activity worksheets 1–3 2. Observe activity process 3. Observe student behavior                                                        | Assessment forms                                        | Teacher Jongruk, Iham Kham, Local wisdom teacher at Chaona School |
| - | <b>Rice Farming</b>                                                                                                                                                                                      | Explain and practice rice seedling methods and rice farming                                                                                         | Research/discussion/explanation/ hands-on seedling practice and different planting methods                                                                                                                     | Observe process and student behavior                                                                                                            | Assessment forms                                        | Teacher Jongruk, Iham Kham, Local wisdom teacher                  |
| 5 | <b>Methods of rice seedling nursery in local area</b> - Broadcast nursery method - Tray nursery method - Transplanting (dam na) - Throw planting (na yon) - Broadcast planting (na wan)                  |                                                                                                                                                     |                                                                                                                                                                                                                |                                                                                                                                                 |                                                         |                                                                   |
| - | <b>Rice Farming (Practice)</b> - Steps in rice farming                                                                                                                                                   | Explain steps in rice farming and practice planting in local areas                                                                                  | Explain/hands-on transplanting and line planting                                                                                                                                                               | Observe process and student behavior                                                                                                            | Assessment forms                                        | Teacher Jongruk, Iham Kham, Local wisdom teacher at Chaona School |

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| - | <b>Post-Farming Cultural Traditions</b> - Buffalo spirit ceremony (Su Khwan Khwai) - Rice destiny ceremony (Sueb Chata Khao)                                                           | Research and explain the steps of the buffalo spirit ceremony and rice destiny ceremony — local wisdom and meaning of ritual materials | Research/explanation/community ritual participation                                                                                                                                                      | Observe process and student behavior, check video clips                                                                                                                                                                   | Assessment forms for behavior and ritual processes           | Teacher Jongruk, Iham Kham, Local wisdom teacher at Chaona School                                                                                       |
| - | <b>Botanical Characteristics of the Rice Plant</b> - Roots - Stem - Leaves - Plant growth - Tillering                                                                                  | Survey, examine, discuss and explain botanical characteristics of the rice plant, its growth and tillering                             | Survey/examine/discuss/explain/hands-on                                                                                                                                                                  | 1. Check poster of rice plant structure 2. Assess scientific process skills 3. Observe student behavior                                                                                                                   | Assessment forms for posters, scientific skills and behavior | Teacher Jongruk, Iham Kham, Local wisdom teacher at Chaona School                                                                                       |
| - | <b>Rice Field Ecosystem</b> - Weeds - Rice pests/insects - Rice diseases - Natural enemies - Benefits of pest animals - Paddy crab - Biological fermentation liquid from banana shoots | Survey and examine rice diseases, insects, pests, natural enemies, weeds in rice fields and benefits of pest animals                   | - Group search/presentation on rice diseases, insects, weeds, natural enemies - Food web game of rice ecosystem - Hands-on survey of rice field ecosystem - Brainstorm approaches to reduce chemical use | 1. Check worksheet on diseases, insects, pests and weeds in rice fields 2. Check video clip on paddy crab and crab-based dishes 3. Check ecosystem poster 4. Assess scientific process skills 5. Observe student behavior | Assessment forms for all above                               | Aj. Phongchai and team to design rice ecosystem game (food chain, food web, pesticide accumulation in ecosystem), RMUT banana shoot fermentation liquid |